

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Conditions for stop sending LFPS during Gen 4 CLx Exit

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Conditions for stop sending LFPS during Gen 4 CLx Exit. Today there is separation between the router that initiate the exit and the router that responds to the request (unlike the retimer behavior)

Benefits as a result of the changes:

Spec clarification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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NA

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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NA

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Actual Change

(a). 4.2.1.6.5.4 Gen 4 Exit Flow from CL1 or CL2 State

To Text:

A USB4 Port may initiate exit from CL1 or CL2 state due to an internal objection or detection of LFPS on its Lane 0 receiver. When initiating exit from CL1 or CL2 state, a USB4 Port shall:

1. Send a Low Frequency Periodic Signaling (LFPS) burst on Lane 0 until all the following two conditions are met:
 - The receiver detected LFPS.
 - After the receiver detected LFPS, the transmitter sent at least 16 LFPS cycles.
 - ~~• If the exit was initiated by detecting LFPS:
 - ~~○ The receiver no longer detects LFPS.~~
 - ~~○ The transmitter sent at least 16 LFPS cycles.~~~~
 - ~~• Else:
 - ~~○ The receiver detected LFPS.~~
 - ~~○ After the receiver detected LFPS, the transmitter sent at least 16 LFPS cycles.~~~~

If exit was initiated by detecting LFPS, the Port may apply a third condition:

 - The receiver no longer detects LFPS.

The Port shall stop sending LFPS within tStopLFPS2 time after the conditions above are met.

If the exit from CL1 or CL2 was initiated by detecting LFPS on the Lane 0 receiver, the Port shall transmit the LFPS on the Lane 0 transmitter within tWarmupCL1 or tWarmupCL2 respectively.

If the receiver did not detect LFPS after tTrainingAbort2 time, the Router shall initiate a Disconnect by driving SBTx to a logical low state for tDisconnectTx.
2. Return to Electrical Idle for tPreData (see Table 3-33).